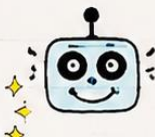




UNIT 1-



Introduction to Artificial Intelligence

1. What is AI ?

Artificial Intelligence (AI) is the simulation of human intelligence in machines that are programmed to **think** and **learn** like humans and **mimic** their actions.

3. Brief History of AI

- **1956** - John McCarthy coined the term "Artificial Intelligence" at Dartmouth Conference.
- **1960s** - Early AI programs were developed (e.g., Logic Theorist, ELIZA).
- **1980s** - Expert Systems became popular.
- **1990s** - Machine Learning techniques grew.
- **2000s to Present** - Deep Learning, Big Data and powerful computing boosted AI applications.

5. Applications of AI

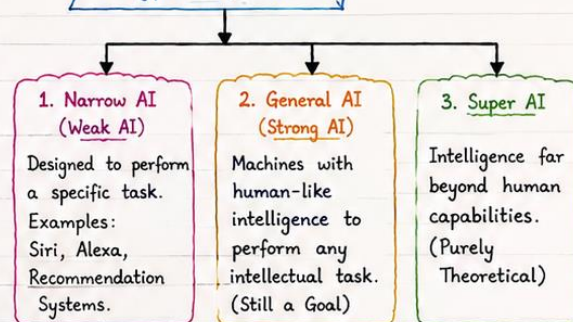
- ★ **Healthcare** - Disease diagnosis, robotic surgery.
- ★ **Education** - Intelligent tutoring systems.
- ★ **Finance** - Fraud detection, algorithmic trading.
- ★ **Transport** - Self-driving cars, traffic prediction.
- ★ **Entertainment** - Movie recommendations, video games.
- ★ **Daily Life** - Virtual assistants, smart homes.



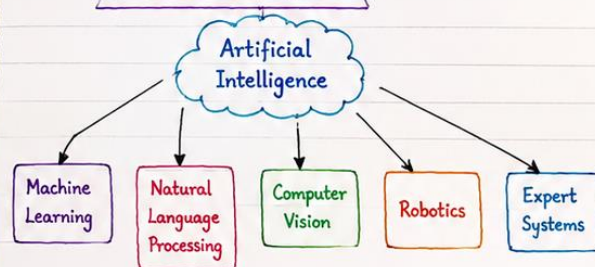
2. Characteristics of AI

- **Intelligence** - ability to learn and solve problems.
- **Learning** - improvement from experience.
- **Reasoning** - ability to make logical decisions.
- **Perception** - understanding the environment through senses.
- **Autonomy** - ability to act without human intervention.

4. Types of AI



6. Branches of AI



★ **Conclusion** : Artificial Intelligence is transforming the world by making machines smarter and life easier. It combines knowledge from multiple fields to create intelligent systems that can benefit humanity in countless ways. ★

Q.2 What are AI problems? Explain with examples.

(7 Marks)

Ans: AI problems are problems that require intelligent behavior to solve. These problems cannot be solved by using simple algorithms and need reasoning, learning or perception.

Examples of AI problems :

- Game playing: Playing games like Chess, Carrom or Go requires decision making, strategy and prediction of opponent moves.
- Natural Language Understanding: Understanding human language, answering questions, translation, etc.
- Speech Recognition: Converting spoken language into text. e.g., voice typing, voice assistants.
- Computer Vision: Understanding and interpreting images or videos. e.g., face recognition, object detection.
- Expert Systems: Systems that give expert level advice in specific domains like medical diagnosis.
- Robotics Navigation: Robots sense the environment and make decisions to move safely and reach goals.
- Planning and Scheduling: Deciding the best sequence of actions to achieve a goal. e.g., travel planning.
- Machine Translation: Translating text from one language to another automatically.
- Autonomous Driving: Self-driving cars that perceive the road, detect obstacles and take decisions.

These problems involve uncertainty, incomplete information and require knowledge and experience to solve them.

Q.4 Write levels of models in AI.

(7 Marks)

Ans: AI models can be classified into 4 levels based on their capabilities. These levels show the evolution of AI from imitating humans to taking rational decisions.

1. Thinking Humanly (Cognitive Modeling): Systems that think like humans by using mental processes such as reasoning, problem solving and learning.
e.g., cognitive architectures, human like problem solvers.
2. Acting Humanly (Turing Test Approach): Systems that act like humans. The goal is to pass the Turing Test, i.e., the machine's behavior is indistinguishable from human behavior.
e.g., chatbots, virtual assistants.
3. Thinking Rationally (Laws of Thought): Systems that think using formal logic and rules of inference to reach correct conclusions.
e.g., theorem provers, logic based expert systems.
4. Acting Rationally (Rational Agent): Systems that act in the best possible way to achieve goals or maximize success based on available information.
e.g., self-driving cars, game playing agents.

These levels represent increasing maturity of AI systems from imitating human thinking to making optimal decisions autonomously.

Q1. Explain AI Problem Characteristics in detail.

[7 Marks]

Ans: An AI problem is a problem that requires reasoning, decision-making, searching or intelligent behavior to obtain a solution. The important characteristics of an AI problem are as follows :

- 1) Decomposability : A complex problem can sometimes be divided into smaller sub-problems.

Example : Playing chess → opening → middle game → end game.

- 2) Nature of Solution Steps : Some problems require a sequence of steps to reach the final solution.

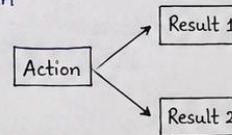
Initial State → Action 1 → Action 2 → Action 3 → Goal State

- 3) Recoverability of Steps : We need to know whether a wrong step can be reversed.

- Recoverable : Chess move can be changed by another move.
- Non-recoverable : A robot damaging an object cannot easily reverse the action.

- 4) Predictability : We consider whether the result of an action can be predicted.

- Predictable environment : Same action → same result.
- Unpredictable environment : Result may vary.



- 5) Quality of Solution : AI may have several possible solutions. The system should select the best or most suitable solution.

- 6) Knowledge Requirement : Some problems require large amount of knowledge.

Example : Symptoms + Medical Knowledge → Reasoning → Possible Disease

- 7) Cost of Solution : The solution may be evaluated based on time, memory, computational resources and accuracy.

Key Points

- Decomposability
- Nature of Steps
- Recoverability
- Predictability
- Quality
- Knowledge Requirement
- Cost of Solution

*

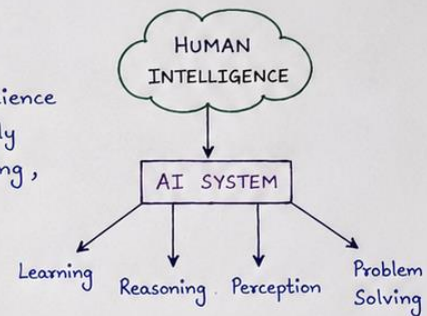
Conclusion : AI problems are generally characterized by their states, actions, goals, knowledge requirements, predictability and solution cost.

Q3. What is Artificial Intelligence? Explain applications of AI.

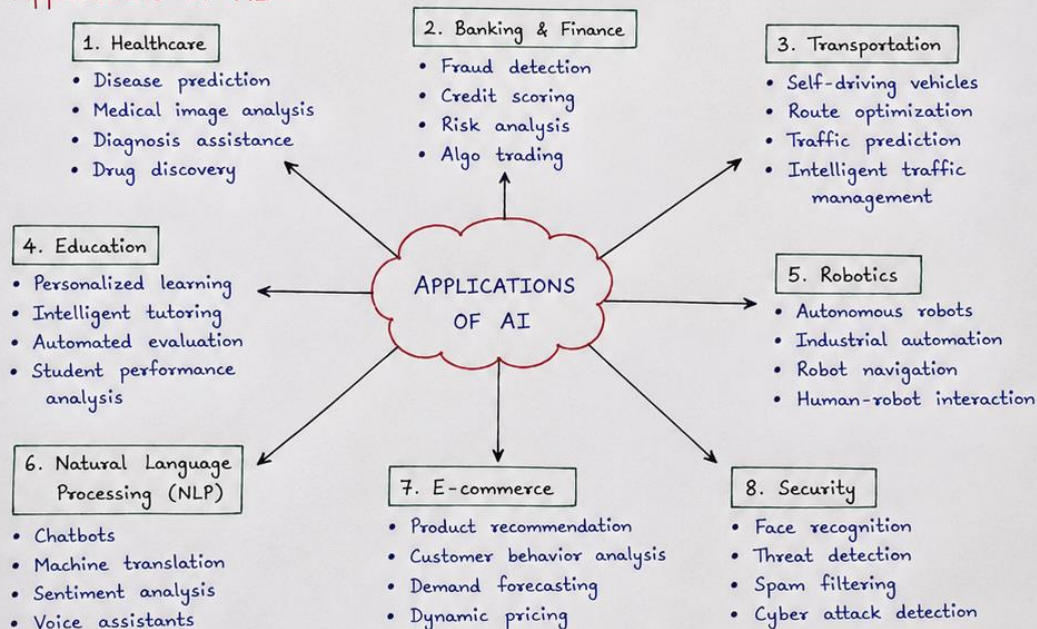
[7 MARKS]

Ans: (1) What is Artificial Intelligence ?

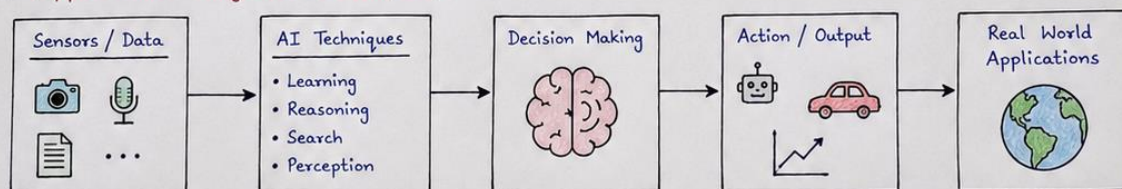
Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence such as learning, reasoning, problem solving, perception and decision making.



(2) Applications of AI :



(3) Application Diagram (Examples)



(4) Conclusion : AI is used in almost every field to make systems intelligent, efficient and accurate. It helps in improving human life and making better decisions.

Q4. Explain the underlying assumptions of Artificial Intelligence.

7 Marks

Ans: The development of AI is based on certain fundamental assumptions about intelligence. These assumptions guide the design and functioning of intelligent systems.

1) Intelligence can be represented :

It is assumed that human knowledge, facts, rules and experience can be represented in a form that a computer can store and manipulate.

Example : Rules like "If fever then disease" can be written in IF-THEN form.

2) Reasoning can be mechanized :

The processes of reasoning and inference can be carried out by computers using logical rules.

Example : From "All humans are mortal" and "Socrates is human", we can infer "Socrates is mortal".

3) Knowledge is transferable :

Knowledge from one person (expert) can be transferred to a machine and used to solve problems.

Example : Expert systems store knowledge of doctors, engineers etc.

4) Problems can be solved by search :

Solutions to a problem can be found by searching through possible alternatives.

Example : Route finding in maps using search algorithms.

5) Machines can learn from experience :

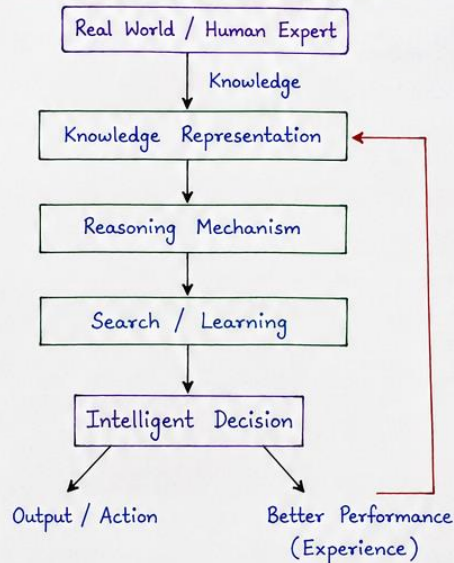
It is assumed that machines can improve their performance by learning from data or past experience.

Example : A spam filter learns from previous emails.

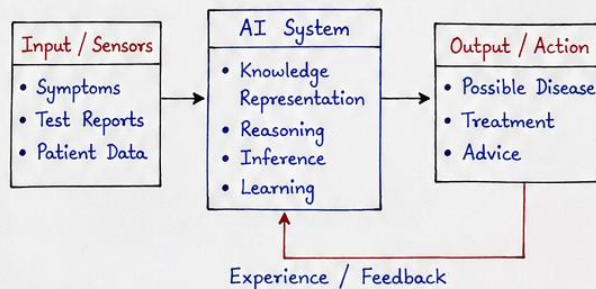
6) Human intelligence is not mysterious :

Human intelligence is based on natural processes such as perception, memory, reasoning etc. and these processes can be simulated.

How these assumptions work together :



Example : Medical Diagnosis System



In Short :

The underlying assumptions of AI are :

1. Intelligence can be represented.
2. Reasoning can be mechanized.
3. Knowledge can be transferred.
4. Problems can be solved by search.
5. Machines can learn from experience.
6. Human intelligence can be simulated.

Conclusion : These assumptions provide the foundation for building intelligent systems that can perceive, reason, learn and act like humans. *

Q.1 What is Artificial Intelligence (AI)? Explain the need of AI. (7 Marks)

Ans: Artificial Intelligence (AI) is a branch of computer science that deals with the creation of intelligent machines that can perform tasks which normally require human intelligence, such as learning, reasoning, problem solving, perception and understanding natural language.

Need of AI:

- **Automation of difficult tasks:** AI helps to automate tasks that are difficult, repetitive or dangerous for humans.
e.g., robots in factories, self-driving cars.
- **Improved accuracy and speed:** AI systems can process large amount of data quickly and with high accuracy.
e.g., medical diagnosis, fraud detection.
- **Solving complex problems:** AI can solve complex problems in science, engineering, finance etc. which are difficult for traditional methods.
- **Learning from experience:** AI systems can learn from data and improve their performance over time.
e.g., recommendation systems, voice assistants.
- **Enhancing human capabilities:** AI acts as a powerful tool to assist humans in decision making and problem solving.
e.g., virtual assistants, expert systems.
- **Better decision making:** AI analyzes large data and provides useful insights for better and faster decisions.
e.g., stock prediction, weather forecasting.

Q.1 What is Artificial Intelligence (AI) ?

Explain the need of AI. (7 Marks)

Ans:

Artificial Intelligence (AI) is the branch of computer science that deals with the creation of intelligent machines that can perform tasks which normally require human intelligence, such as learning, reasoning, problem solving, perception and understanding natural language.

Need of AI:

- To automate tasks that are difficult, repetitive or dangerous for humans.
- To improve decision making with accuracy and speed.
- To solve complex problems in science, medicine, engineering, finance etc.
- To develop systems that can learn from experience and adapt to new situations.
- To enhance human capabilities and improve quality of life.

Q.2 What are AI problems ? Explain with examples. (7 Marks)

Ans:

AI problems are problems that require intelligent behavior to solve. These problems cannot be solved by using simple algorithms and need reasoning, learning or perception.

Examples of AI problems:

- Game playing (e.g., Chess)
- Natural Language Understanding
- Speech Recognition
- Computer Vision (Image Recognition)
- Expert Systems (Medical Diagnosis)
- Robotics Navigation
- Planning and Scheduling
- Machine Translation
- Handwriting Recognition
- Autonomous driving

These problems involve uncertainty, incomplete information and require knowledge and experience to solve them.

Q.3 Explain AI techniques. (7 Marks)

Ans: AI techniques are methods or approaches used to build intelligent systems.

Some important AI techniques are:

1. Problem Solving – Searching techniques, heuristics, means-ends analysis.
2. Knowledge Representation – Using logic, rules, semantic networks, frames.
3. Reasoning – Deductive, Inductive and Abductive reasoning.
4. Learning – Machine learning, neural networks.
5. Natural Language Processing – Parsing, translation, speech understanding.
6. Perception – Computer vision, pattern recognition.
7. Expert Systems – Rule-based systems using domain knowledge.
8. Robotics – Sense, plan and act in real world.

Q.4 Write levels of models in AI. (7 Marks)

Ans: AI models can be classified into 4 levels based on their capabilities.

1. Thinking Humanly (Cognitive Modeling)
 - Systems that think like humans.
2. Acting Humanly (Turing Test Approach)
 - Systems that act like humans.
3. Thinking Rationally (Laws of Thought)
 - Systems that think using logic and rules.
4. Acting Rationally (Rational Agent)
 - Systems that act in the best possible way to achieve goals.

These levels show the evolution of AI from imitating humans to taking rational decisions.

Q.5 What are the criteria for success of AI ? (7 Marks)

Ans: The success of AI can be measured based on the following criteria:

1. Performance – Accuracy, speed and efficiency of the system.
2. Reliability – Consistency of results in different situations.
3. Scalability – Ability to handle large amount of data and complex problems.
4. Adaptability – Ability to learn and adapt to new environments.
5. Usability – Easy to use and understand by humans.
6. Cost Effectiveness – Achieve goals with minimum cost and resources.
7. Robustness – System should work well even with incomplete or noisy data.

Q.6 Write applications of Artificial Intelligence. (7 Marks)

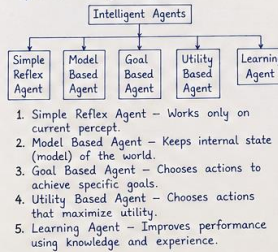
Ans: Applications of AI:

- Expert Systems – Medical diagnosis, financial planning.
- Natural Language Processing – Chatbots, machine translation.
- Computer Vision – Image recognition, face detection.
- Speech Recognition – Voice assistants, speech to text.
- Robotics – Industrial robots, autonomous vehicles.
- Game Playing – Chess, Go, video games.
- Recommendation Systems – YouTube, Netflix, Amazon.
- Healthcare – Disease prediction, drug discovery.
- Finance – Fraud detection, stock prediction.

Q.7 What is an intelligent agent ? Explain types of intelligent agents. (7 Marks)

Ans: An intelligent agent is an entity that perceives its environment through sensors and acts upon that environment through actuators to achieve goals.

Types of Intelligent Agents:



Q.8 Write characteristics of AI systems. (7 Marks)

Ans: Characteristics of AI systems:

- Intelligence – Ability to learn, reason and solve problems.
- Autonomy – Ability to operate without human intervention.
- Adaptability – Can adapt to new situations.
- Learning Ability – Improves performance from past experience.
- Reasoning – Draws conclusions from facts and rules.
- Perception – Can perceive environment using sensors.
- Goal Oriented – Works to achieve specific goals.
- Uncertainty Handling – Can handle incomplete or uncertain information.

Q.3 Explain AI techniques.

(7 Marks)

Ans: AI techniques are methods or approaches used to build intelligent systems. Some important AI techniques are :

1. **Problem Solving**: It is the most basic AI technique. The system searches for a sequence of actions to reach a goal.
e.g., 8-puzzle problem, path finding.
2. **Knowledge Representation**: Knowledge about the world is represented in a form that machines can understand and use.
e.g., using logic, rules, semantic networks, frames.
3. **Reasoning**: It is the process of deriving new facts or conclusions from known facts. Types - Deductive, Inductive, Abductive.
e.g., medical diagnosis using rules.
4. **Machine Learning**: Systems learn from data and improve their performance without being explicitly programmed.
e.g., spam detection, recommendation systems.
5. **Natural Language Processing (NLP)**: It enables computers to understand, interpret and generate human language.
e.g., chatbots, machine translation.
6. **Perception**: It allows machines to interpret sensory data like images, speech, sound, etc.
e.g., image recognition, speech recognition.
7. **Expert Systems**: It uses knowledge and inference rules of human experts to solve complex problems in a specific domain.
e.g., medical diagnosis, fault detection.
8. **Robotics**: It involves the design of intelligent robots that can sense, plan and act in the real world.
e.g., industrial robots, autonomous robots.

These techniques are used individually or in combination to create intelligent applications that solve real-world problems.